

Two-Station Machining Cell Control System



Simulation-Based
PLC-Equivalent
Electrical Control Design



A microcontroller-based system that
sequences, interlocks and protects
a two-station machining and
material-handling cell.

KEY TECHNOLOGIES



STM32F103C8T6
Microcontroller



16x4 LCD
Status Display



Industrial Relay
Output Switching



Emergency Stop
Safety Input



Push Button
User Input



Potentiometer
Overload Simulation



こうじょうのかだい Industrial Problem & Motivation



In modern manufacturing, small errors in sequence or protection can lead to **big losses**.



The Need



Manufacturing cells need a reliable control system that **sequences, protects, monitors, and ensures quality** — all in real time.



Safe
Operation



Protected
Equipment



Consistent
Quality



Real-Time
Visibility



This project addresses these challenges by designing a robust, protection-integrated control system for a two-station machining cell.



SYSTEM OVERVIEW



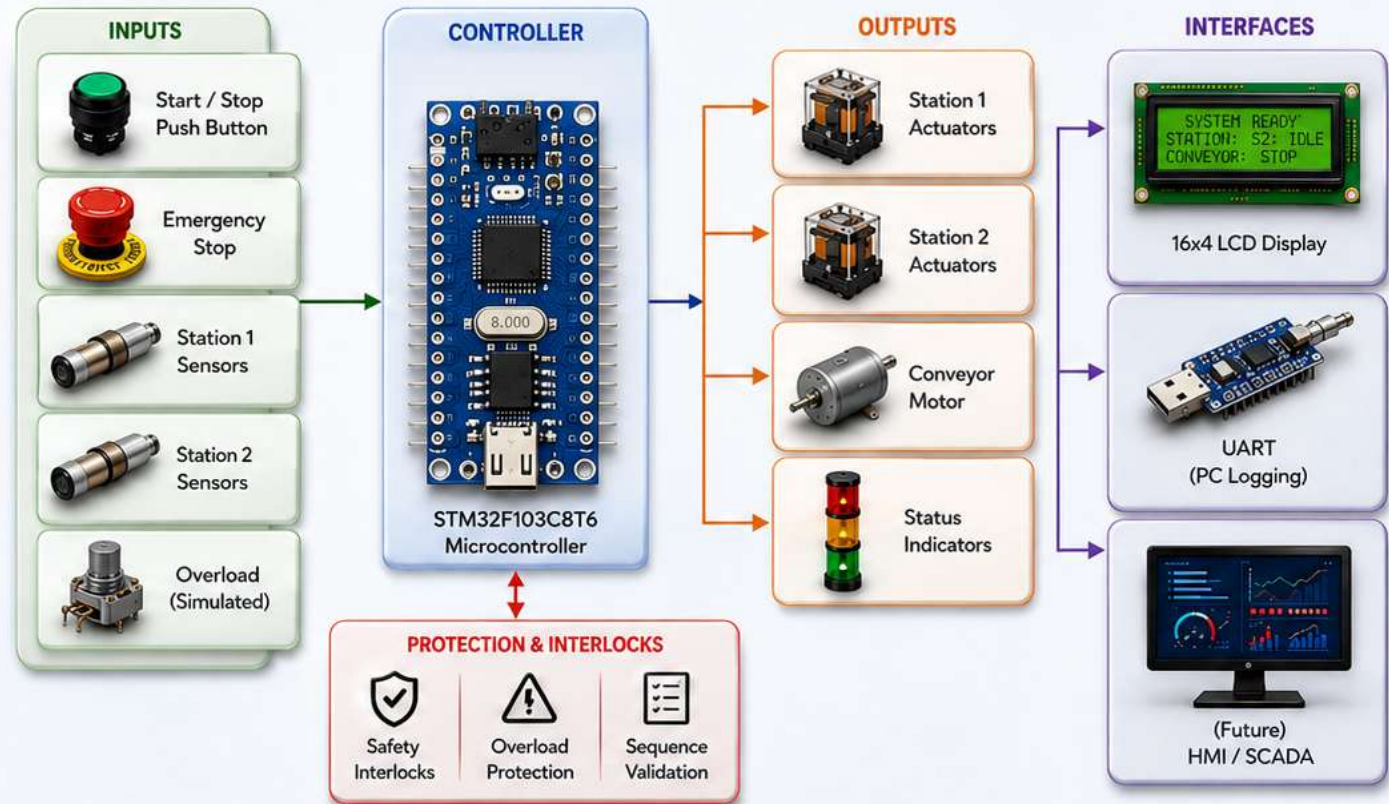
What This System Does

- ✓ Controls the sequence of **two machining stations** and material handling.
- ✓ Ensures **safety interlocks** and **overload protection**.
- ✓ Provides **real-time feedback** and **event logging**.



A reliable control solution that ensures safe operation, protects equipment and operators, and maximizes productivity.

SYSTEM ARCHITECTURE



KEY BENEFITS



Safe Operation

Minimizes risk to operators and equipment



Accurate Sequencing

Ensures correct order of operations and smooth flow



Equipment Protection

Overload and fault protection prevents unexpected downtime



Real-Time Visibility

Live status and alarms for quick decision making



Event Logging

Records important events for analysis and traceability



Scalable Solution

Easy to expand to more stations and integrate with higher systems

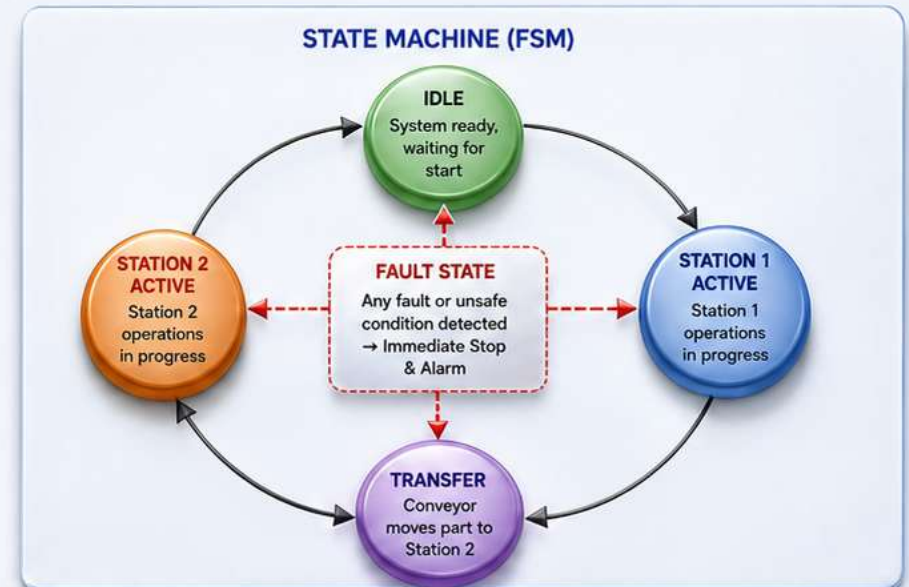
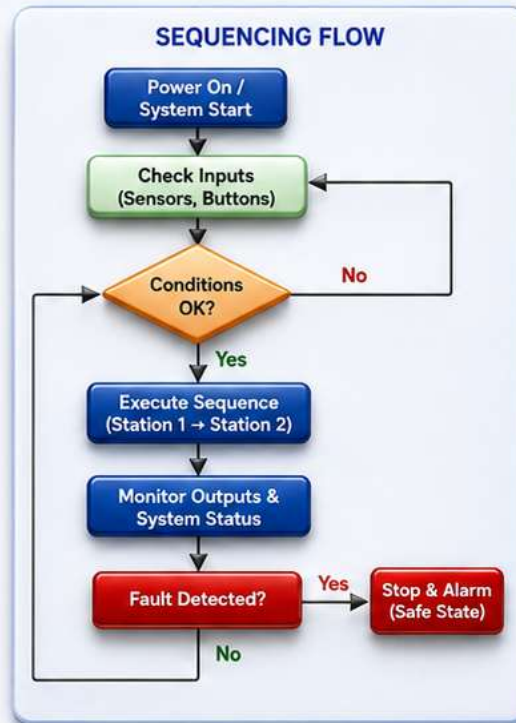
CONTROL LOGIC

Smart Sequencing & Safe Interlocks



HOW IT WORKS

The controller continuously checks, inputs, validates conditions and executes the correct sequence while enforcing all safety rules.



CONTROL & PROTECTION FEATURES



RELAY INTERLOCK
Prevents uncommanded motion and backfeed.



FUSE PROTECTION
Shields circuits from overcurrent and short circuit.



SMPS ISOLATION
Provides stable, isolated DC power to the system.



OPTO-ISOLATION
Isolates control and power circuits for safe operation.



TERMINAL BLOCKS
Secure and organized wiring connections.



MCB PROTECTION
Protects the system from overload and faults.

SENSOR & INTERLOCK DESIGN

Reliable Detection. Safe Operation.

SENSOR STRATEGY



Locator Check
Verify proximity at key points.



Limit Switches
Confirm mold presence.



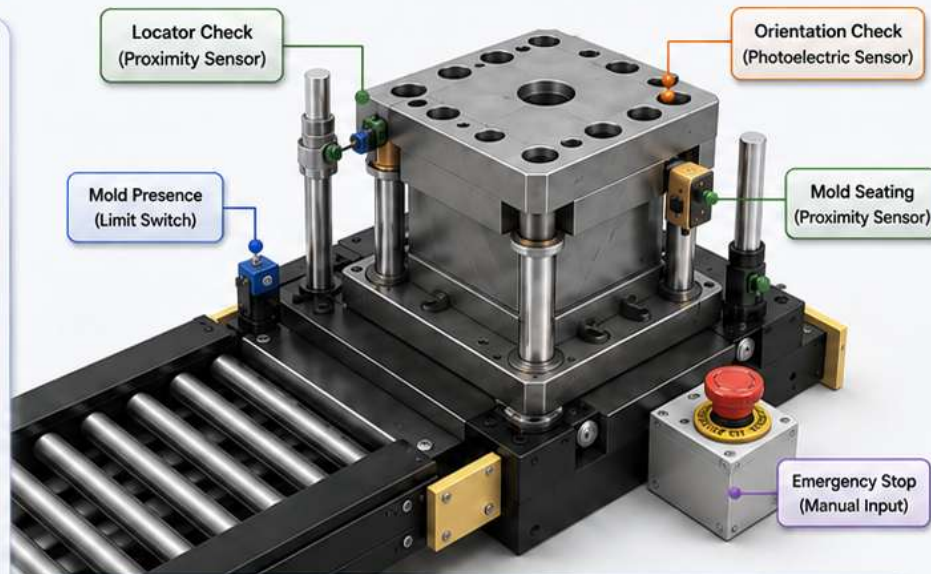
Inductive Sensors
Ensure correct mold seating.



Photoelectric Sensors
Validate orientation & locator engagement.



Manual Inputs
Start / Stop & Emergency Stop.



Multiple sensors work together to detect presence, seating and orientation.

Only when all conditions are satisfied, the system allows the cycle to start.



SAFETY INTERLOCK LOGIC



All critical sensors must be in correct state.



If any sensor detects an error, the system enters **FAULT STATE**.



Machine start is permitted only when all conditions are OK.



Prevents mis-seating, wrong orientation and equipment damage.



DESIGN BENEFITS



High Reliability



Reduced Downtime



Enhanced Safety



Easy to Maintain



Simple & Robust



Scalable for Future

KEY HARDWARE USED



LIMIT SWITCH



INDUCTIVE SENSOR



PHOTOELECTRIC SENSOR



EMERGENCY STOP BUTTON



RELAY



TERMINAL BLOCKS



MCB

SIMULATION & RESULTS

IMPLEMENTATION FLOW



SIMULATED PROTOTYPE (Proteus 8 Professional)

The system is designed and tested in Proteus 8 Professional to validate logic, interlocks and output responses.



TEST SCENARIOS

- ✓ Correct mold seating & orientation
- ✓ Missing mold / incorrect position
- ✓ Wrong orientation detected
- ✓ Sensor failure / disconnection
- ✓ Emergency stop activation

1



Inputs Activated

Sensors & buttons provide status

2



Controller Processing

Conditions are checked as per logic rules

3



Safety Interlocks

All conditions must be OK to proceed

4



Outputs Triggered

Actuators run & status indicators update

5



Start Permit Issued

Machine starts only when system is safe

SIMULATION RESULTS (examples)

ALL CONDITIONS OK



Green LED ON
Buzzer OFF
Relay ON

START PERMITTED

MOLD NOT SEATED



Red LED ON
Buzzer ON
Relay OFF

START BLOCKED

WRONG ORIENTATION



Yellow LED ON
Buzzer ON
Relay OFF

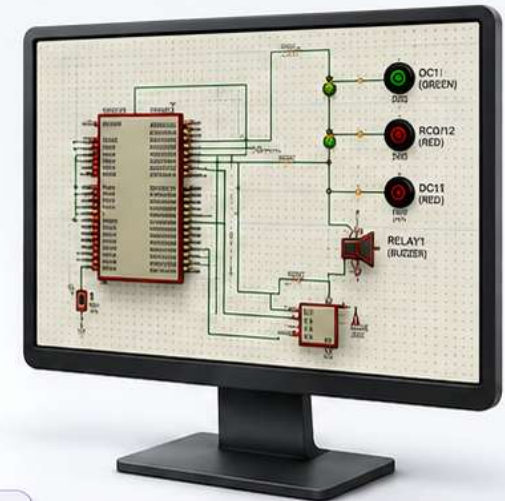
START BLOCKED

EMERGENCY STOP



All Outputs OFF
Buzzer ON
System Halted

IMMEDIATE STOP



Proteus 8 Professional Screenshot



HIGH ACCURACY

Accurate detection of seating & orientation in all test cases.



ZERO FALSE START

Prevents machine start in every unsafe condition.



FAST RESPONSE

Quick detection & interlock action within milliseconds.



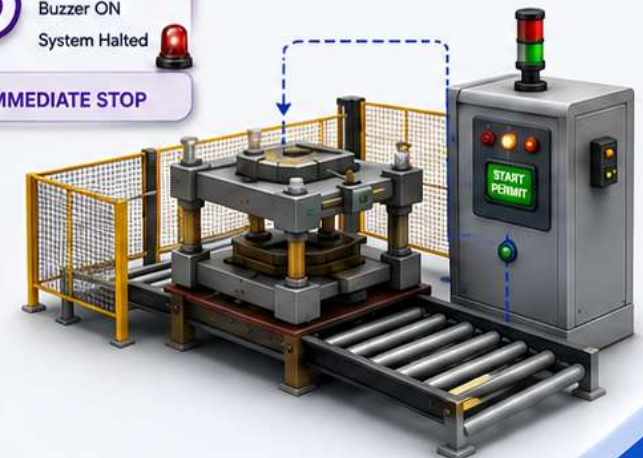
RELIABLE OPERATION

Stable performance in continuous & repeated test cycles.



SAFETY FIRST

Ensures operator safety, mold protection & equipment reliability.



The system is fully simulated in Proteus 8 Professional and verified for safety, reliability and correct operation.

APPLICATIONS & FUTURE SCOPE

KEY ADVANTAGES



Prevents Damage & Downtime

Stops the machine if mold is not seated or oriented correctly.



Improves Safety

Ensures operator and equipment safety through reliable interlocks.



Ensures Quality

Guarantees proper mold positioning for consistent product quality.



Saves Time & Cost

Reduces manual checking and prevents production delays.



Easy to Integrate & Scale

Simple design that can be integrated into existing lines and expanded easily.

REAL-WORLD APPLICATIONS



Injection Molding Machines

Ensure mold is seated and oriented before cycle start.



Die Casting Machines

Verify die position and locking before casting.



Press & Stamping Lines

Confirm tool alignment and presence for safe operation.



Automated Production Cells

Safety gate for starting sequence in automated environments.



SKILLS & KNOWLEDGE DEMONSTRATED



Microcontroller Programming



Sensor Selection & Placement



Safety & Interlock Design



Circuit Design & Simulation



System Thinking



Problem Solving & Analysis



ありがとうございます

